

## **Laboratory No. 1 – Electrochemistry Unplugged: Make Your Own Potentiostat**

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### **1. Workshop goals**

By the end of this laboratory you will have assembled a minimal three-electrode potentiostat from four op-amp stages on a breadboard, uploaded its control firmware, and recorded a cyclic voltammogram of a test system, while understanding what each stage of the circuit does and why it is needed.

### **2. Theoretical introduction**

A solderless **breadboard** connects components without soldering: the two outer rows on each side are power rails running the full length of the board, while the central area is split into short vertical strips of five holes that are joined internally.

Chip stay at the breadboard' centre gap, like our LM324, lands one row of legs in each half.

The **digital multimeter** is your main diagnostic tool today: DC-voltage mode checks that a node sits at the voltage you expect, while continuity/resistance mode confirms whether two points are (or are not) connected, which is the fastest way to catch a wiring mistake or an accidental bridge to ground.

Of the components: a **resistor** limits current and sets gain (value in ohms); a **capacitor** stores charge and here is used mainly to stabilise the feedback loops and filter noise, not to store energy; an operational amplifier (**op-amp**) is a very high-gain differential amplifier that, once wired with negative feedback, forces its two inputs to the same voltage. The LM324 packs four independent op-amps.

The microcontroller (**XIAO SAMD21**) is a digital device: it only produces and reads discrete numbers, not continuous voltages. A Digital-to-Analog Converter (**DAC**) turns a number into an analog voltage. An Analog-to-Digital Converter (**ADC**) does the reverse: it samples an analog voltage and reports it as a number the microcontroller can log. Because the microcontroller has only a single positive supply and cannot produce negative voltages, the whole analog circuit is built around a virtual ground: an artificial “zero” fixed at the midpoint of the supply ( $\approx 1.65$  V here) instead of the real 0 V rail. Every potential in the circuit is then meant relative to this point, so a cell potential that should swing positive and negative around zero can still be handled by op-amps that only ever see voltages between 0 V and the supply rail. Verify with the multimeter.

**The minimum potentiostat: VF, CA, TIA.** A three-electrode potentiostat needs only three functional building blocks, each a single op-amp:

- **Voltage Follower (VF):** a unity-gain buffer: whatever voltage is applied to its input appears at its output, but the output can supply real current while the input draws almost none. We use two: one to generate a clean, low-impedance virtual ground from a resistor divider, and one to “read” the reference electrode (RE) without loading it, since a real reference electrode must carry essentially zero current or its own potential drifts.
- **Control Amplifier (CA):** the feedback engine of the potentiostat: it compares the potential you want to apply (from the DAC) with the potential actually measured at RE (through the buffer above), and drives the counter electrode (CE) up or down until the two agree. This is the loop that lets you set the cell's potential.
- **Transimpedance Amplifier (TIA):** how current, not voltage, is measured: it holds the working electrode (WE) at the virtual ground and converts the tiny current flowing through it into a voltage using a single feedback resistor (current = voltage  $\div$  resistance). This is the signal the ADC ultimately digitises.

Chained together — DAC → CA → CE → cell → RE → VF → back to CA, closing the potential loop; and WE → TIA → ADC, reading the current — these four op-amp stages are the complete, minimal architecture of any potentiostat (Figure 1).

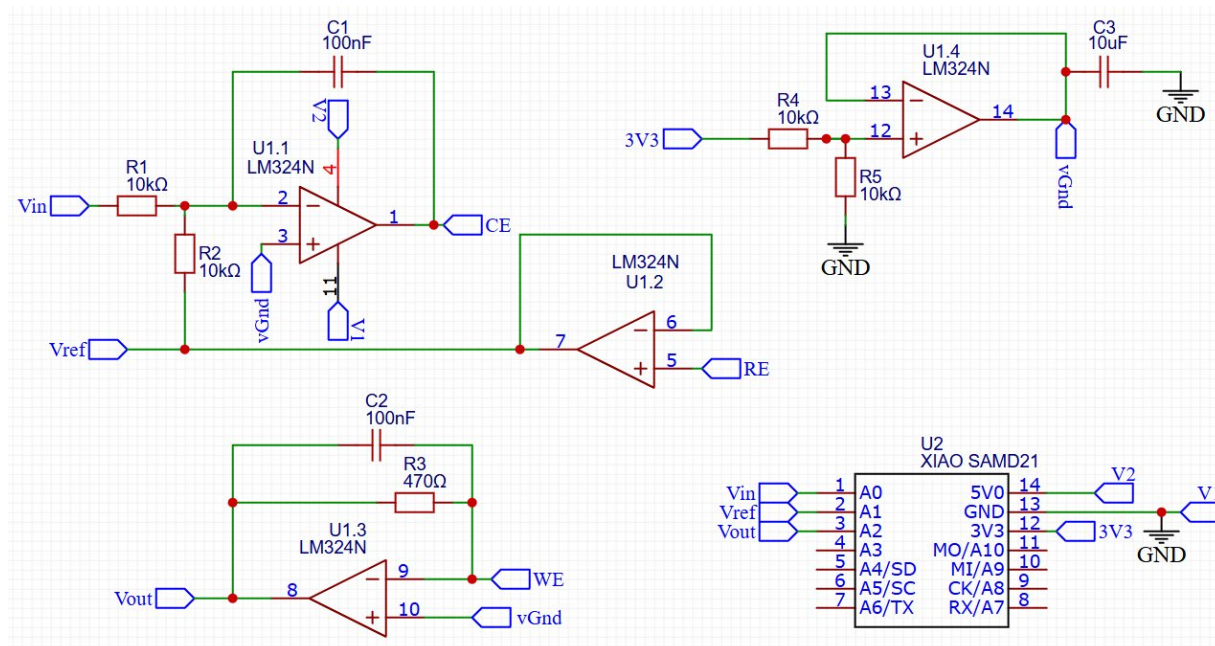


Figure 1. Signal-flow schematic of the minimal 4-op-amp potentiostat (LM324) used in this laboratory. The full component-level schematic, parts list and firmware are provided in the accompanying GitHub repository.

### 3. Laboratory setup

Each station needs the following. The full component-level schematic, BOM and firmware referenced below are in the GitHub repository accompanying this laboratory.

- Solderless breadboard and jumper wires
- Digital multimeter
- Seeeduino XIAO SAMD21 with USB-C cable, and a laptop with the Arduino IDE and the SAMD21 board package installed
- One LM324 quad operational amplifier (DIP-14), plus one spare
- Resistors: four 10 kΩ (R1, R2, R4, R5) and one 470Ω (R3, sets the current-to-voltage gain of the TIA)
- Capacitors: two 100 nF (C1, C2) and one 10 μF (C3)
- **Quick electronic check:** one 10 kΩ resistor as a “dummy cell” (wired between WE and CE, with RE shorted to CE)
- **Real electrochemistry:** a three-electrode cell with glassy-carbon or Pt working electrode, Ag/AgCl reference electrode, Pt-wire counter electrode, and ≈5 mM  $K_3[Fe(CN)_6]$  in 0.1 M KCl supporting electrolyte

### 4. Execution of the laboratory

Work in pairs. Follow the schematic (Figure 1 and the GitHub repository) at each step.

- 1) Identify the four op-amp sections inside the single LM324 package: pins 1–3 are the control amplifier (CA), pins 5–7 the RE buffer, pins 8–10 the TIA, and pins 12–14 the virtual-ground buffer.
- 2) Place the LM324 straddling the centre gap of the breadboard. Wire pin 4 (V+) to the XIAO’s 5 V pin and pin 11 (V–) to GND.

- 3) Build the virtual-ground generator: R4 (10 k $\Omega$ ) from 3V3 to a node, R5 (10 k $\Omega$ ) from that node to GND, node into pin 12, buffer output on pin 14.
- 4) **Check before continuing:** measure pin 14 (and pin 12) with the multimeter — it must read close to half the supply,  $\approx 1.65$  V. If it does not, stop and look for an accidental short to pin 11 (GND) before wiring anything else.
- 5) Build the reference-electrode buffer (pins 5–7): the RE wire into pin 5, pin 7 wired back to pin 6 (unity-gain feedback).
- 6) Build the control amplifier (pins 1–3): R1 from XIAO A0 into pin 2, R2 from the RE-buffer output (pin 7) into pin 2, C1 as feedback from pin 1 back to pin 2, and pin 3 tied to the virtual-ground bus (pin 14). Pin 1 is CE.
- 7) Build the transimpedance amplifier (pins 8–10): the WE wire into pin 9, R3 (1.45 k $\Omega$ ) in parallel with C2 as feedback from pin 8 back to pin 9, and pin 10 tied to the virtual-ground bus. Pin 8 (VOUT) goes to XIAO A2; pin 7 (VREF) also goes to XIAO A1.
- 8) Connect the test system: for a quick check, wire a single 10 k $\Omega$  resistor between WE and CE and short RE to CE; for the real measurement, connect WE/RE/CE to the three-electrode cell containing the  $K_3[Fe(CN)_6]$  solution.
- 9) **Download the firmware** from the GitHub repository, open it in the Arduino IDE, select “Seeeduino XIAO” as the board, and upload it.
- 10) Open the Serial Monitor at 115200 baud and press Enter to start a cyclic-voltammetry scan; the board streams time, set potential, measured cell voltage and current as CSV.
- 11) Paste the CSV output into a spreadsheet and plot current vs. set potential. With the 10 k $\Omega$  dummy resistor you should see a straight line through the origin (Ohm’s law); with the ferricyanide cell you should see the characteristic redox current peaks of a cyclic voltammogram.

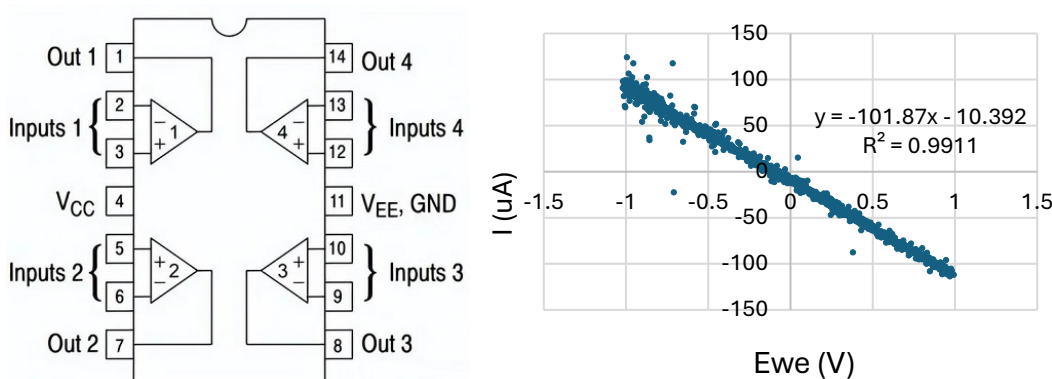


Figure 2. LM324 op-amp and a 10k resistor